Yusheng Liu

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Department of Materials Science and Engineering,
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EDUCATION AND TRAINING

University of Illinois Urbana-Champaign (UIUC) — Illinois, USA
PhD candidate in the department of Materials Science and Engineering — Aug 2021–Present

University of Science and Technology of China (USTC) — Anhui, China
Bachelor of Science (**Hons.**) in Chemistry, School of Chemistry and Materials Science — Aug 2016 – Jun 2020

Princeton University (Department of Molecular Biology) — New Jersey, USA
International Student Internship Program (ISIP) — Jun 2019 – Aug 2019

SELECTED AWARDS AND HONORS

- CCIL Graduate Cancer Scholarship Award — Nov 2023
- Outstanding Graduate Award for Top Performance — Jun 2020
- Guo Moruo Scholarship (Top 1%, the highest USTC honor for undergraduates) — Oct 2019
- Tang Lixin Scholarship (Top 5%) — Oct 2019
- Overseas Research Scholarship for Outstanding Undergraduates (Top 10%) — Jun 2019

PUBLICATIONS

- **Liu, Y.**, Zhou, J., Wang, Y., Baskaran, D., Wang, H. Unnatural Lipids for Simultaneous mRNA Delivery and Metabolic Cell Labeling. **Biomaterials Science** 2024, 12(16), 4170-4180.
- Bo, Y., Wang, Y., Han, J., Bhatta, R., **Liu, Y**., Baskaran, D., Zhou. J., Wang, H. Primary Adipocytes as Targetable Drug Depot to Prevent Post-Surgical Cancer Recurrence. **Materials Today Bio** 2024, 25, 101020.
- **Liu, Y.**, & Wang, H. Biomarkers and targeted therapy for cancer stem cells. **Trends in Pharmacological Sciences** 2023, 45, 56-66.
- **Liu, Y**., Han, J., Bo, Y., Bhatta, R., & Wang, H. Targeted delivery of liposomal chemoimmunotherapy for cancer treatment. **Frontiers in Immunology** 2022, 13, 6031.
- Han, J., Bhatta, R., **Liu, Y.**, Bo, Y., Elosegui-Artola, A., & Wang, H. Metabolic glycan labeling immobilizes dendritic cell membrane and enhances antitumor efficacy of dendritic cell vaccine. **Nature Communications** 2023, 14 (1), 5049.
- Bhatta, R., Han, J., **Liu, Y.**, Bo, Y., Lee, D., Zhou, J., ... & Wang, H. Metabolic tagging of extracellular vesicles and development of enhanced extracellular vesicle based cancer vaccines. **Nature Communications** 2023, 14 (1), 8047.
- Bhatta, R., Han, J., **Liu, Y.**, Bo, Y., & Wang, H. T cell-responsive macroporous hydrogels for in situ T cell expansion and enhanced antitumor efficacy. **Biomaterials** 2023, 293, 121972.
- Han, J., Bhatta, R., **Liu, Y.**, Bo, Y., & Wang, H. In Situ Dendritic Cell Recruitment and T Cell Activation for Cancer Immunotherapy. **Frontiers in Pharmacology** 2022, 13, 954955.
- Han, J., Park, J., Bhatta, R., **Liu, Y.**, Bo, Y., Zhou, J., & Wang, H. A double crosslinking adhesion mechanism for developing tough hydrogel adhesives. **Acta Biomaterialia** 2022, 150, 199-210.
- Wang, W., Hao, Y., **Liu, Y.**, Li, R., Huang, D. B., & Pan, Y. Y. Nanomedicine in lung cancer: Current states of overcoming drug resistance and improving cancer immunotherapy. Wiley Interdisciplinary Reviews: Nanomedicine and Nanobiotechnology 2021, 13(1), e1654.

In Revision/Under Review

- **Yusheng Liu**, Yizun Wang, Yueji Wang, Jiadiao Zhou, Dhyanesh Baskaran, Joonsu Han, Rimsha Bhatta, Daniel Nguyen, Fan Lam, Hua Wang*. In Vivo Metabolic Tagging and Targeting of Red Blood Cells.
- **Yusheng Liu**, Jiadiao Zhou, Yueji Wang, Daniel Nguyen, Dhyanesh Baskaran, Yuan Liu, Hua Wang*. RENBP Inhibition Amplifies Metabolic Labeling Efficiency of Unnatural Mannosamine in Antigen

Presenting Cells in Vitro and in Vivo. PNAS. Submitted.

- Joonsu Han, Abhisek Dwivedy, Tianrui Xue, Rimsha Bhatta**, Yusheng Liu,** Yang Bo**,** Jiadiao Zhou**,** Yueji Wang**,** Joseph Irudayaraj, Shuming Nie, Xing Wang, Hua Wang. Self-Adjuvanting α-Helical Polypeptides for Potent mRNA Cancer Vaccines. Nature Communications. Under Revision.
- Rimsha Bhatta**,** Joonsu Han**, Yusheng Liu,** Yang Bo**,** Hua Wang. Extracellular Vesicle Hydrogels with Tunable Viscoelasticity and Injectability for Depot Vaccines. Nature Communications. Under Revision.
- Yang Bo**,** Yueji Wang, Nilmani Singh, Joonsu Han**,** Jingyi Zhou**,** David Zhou**,** Rimsha Bhatta**, Yusheng Liu,** Dhyanesh Baskaran**,** Shanny Hsuan Kuo, Gee W. Lau, Huimin Zhao, Hua Wang*. Universal Transplant-Specific CAR Regulatory T Cells Prolong Transplant Survival. Under Revision.

PATENTS

- Wang H, **Liu Y**. Metabolic Tagging and Targeting of Red Blood Cells. U.S. Provisional Patent Application No. 63/402,413

PRESENTAIONS

Oral Presentations

- "A Facile Strategy for Long-acting Drug Production via Metabolic Glycoengineering of Erythrocytes"
  2023 BMES Annual Meeting | Seattle Convention Center | Seattle, WA
- "Dual-Functional Azido-Lipids for Efficient Gene Delivery and Membrane Tagging"
  2023 Fall Soft Material Seminar | Room 100, Materials Science and Engineering Building | Urbana, IL
- "Surface GlycoRNA on Dendritic Cells: A Hidden Regulator of T Cell Activation"
  2024 Fall Soft Material Seminar | Room 100, Materials Science and Engineering Building | Urbana, IL

Poster Presentations

- "T cell-responsive Macroporous Hydrogel for in situ T Cell Expansion"
  2022 BMES Annual Meeting | Henry B. Gonzalez Convention Center | San Antonio, TX
- "In vivo Metabolic Tagging and Targeting of Circulating Red Blood Cells "
  2024 Chicago BioEngineering Conference| Chan Zuckerberg Biohub Chicago | Chicago, IL
- "Unnatural Lipids for Simultaneous mRNA Delivery and Metabolic Cell Labeling"
  2024 BMES Annual Meeting| The Baltimore Convention Center | Baltimore, MD

RESEARCH EXPERIENCE

University of Illinois at Urbana-Champaign IL, USA Aug 2021-Present

I) In vivo metabolic glycoengineering of red blood cells (RBCs) for biological application

- Tested the metabolic labeling of RBCs both in vitro and in vivo.
- First time discover the long-term in vivo labeling and cargo targeting of RBCs.
- Conduct blood vessel fluorescence imaging with RBCs labeling technique.
- Successfully image brain blood vessel with MRI imaging over 10 days.
- Controlled blood sugar level in T1D mouse model with RBCs labeling technique.

II) Azido-choline derivative for metabolic lipids labeling and mRNA delivery

- Synthesize and test the labeling of azido-choline with different cell line.
- Synthesize azido-choline analogous and test the labeling and mRNA delivery efficiency.
- Design the chemical structure of azido-choline for UV responsive cancer cell labeling.

III) Lipids coated nanogel for intracellular delivery via membrane fusion for sustained sugar delivery

- Fabricated nano-sized hydrogel via emulsion polymerization and successfully coated with lipids.
- Tested the membrane fusion with cancer cell line for intracellular delivery.

IV) Targeted delivery of liposomal chemoimmunotherapy for cancer treatment

- Fabricate DBCO-liposome and test its targeting effect for azido labeled cancer cell.
- DBCO-liposome encapsulate chemo-drug and adjuvant shows improvement in the immune cells.

University of Science and Technology of China Anhui, China Nov 2019 – Jun 2021

I) Cancer cell antigen-based colloidal gel for cancer immunotherapy

- Fabricated different composition of the colloidal gel and obtained suitable properties.
- Obtained the colloidal gel with ideal properties and tested the gel in the 4T1 cancer model.

II) Engineered red blood cell for type 1 diabetes(T1D) treatment

- Optimized the protocol to conjugate nanoparticle with the red blood cell membrane via click chemistry.

- Conducted in vivo study with NOD mouse model with engineered red blood cells.

Princeton University New Jersey, USA Jun 2019 – Aug 2019

I) The structure of membrane protein A and B

- Purified membrane protein A and B from the recombinant plasmids with HEK293 expression system.

II) The structure of protein C in Nanodisc

- Purified mutated membrane protein C from the mutated plasmid constructed via overlap PCR.
- Synthesized Nanodisc containing protein C and obtained its structure by Cryo-EM with 3D reconstruction.

RELEVANT SKILLS

Chemistry: Organic and inorganic synthesis, nanoparticle synthesis, NMR, MS, HPLC, DLS, SEC(GPC).
Molecular Biology: PCR, plasmid reconstruction, protein expression, protein purification, western blot
Cell Biology: Flow cytometry, stable cell line construction, ex vivo cell culture.
Animal study: Cancer model construction, T1D model construction, mouse surgery.
Computer: C / C++, Origin, Snapgene, Graphpad, Flowjo, MestReNova

TEACHING & MENTORING

1. TA for MSE474 Biomaterials and Nanomedicine, Spring 2023
2. TA for MSE457/CHEM480 Polymer Chemistry, Fall 2022
3. Have mentored three undergraduate researchers since Fall 2021